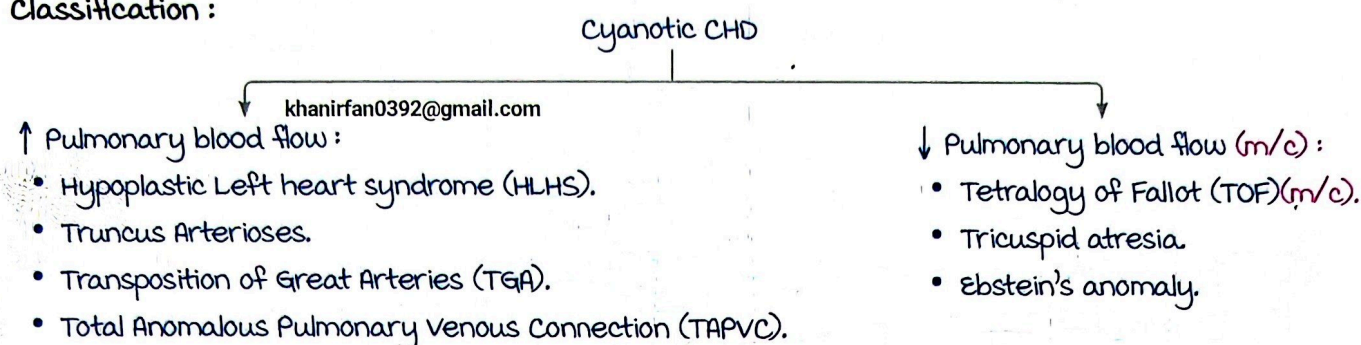


# CYANOTIC CONGENITAL HEART DEFECTS

----- Active space -----

Cyanotic CHD are rarely seen compared to acyanotic CHD.

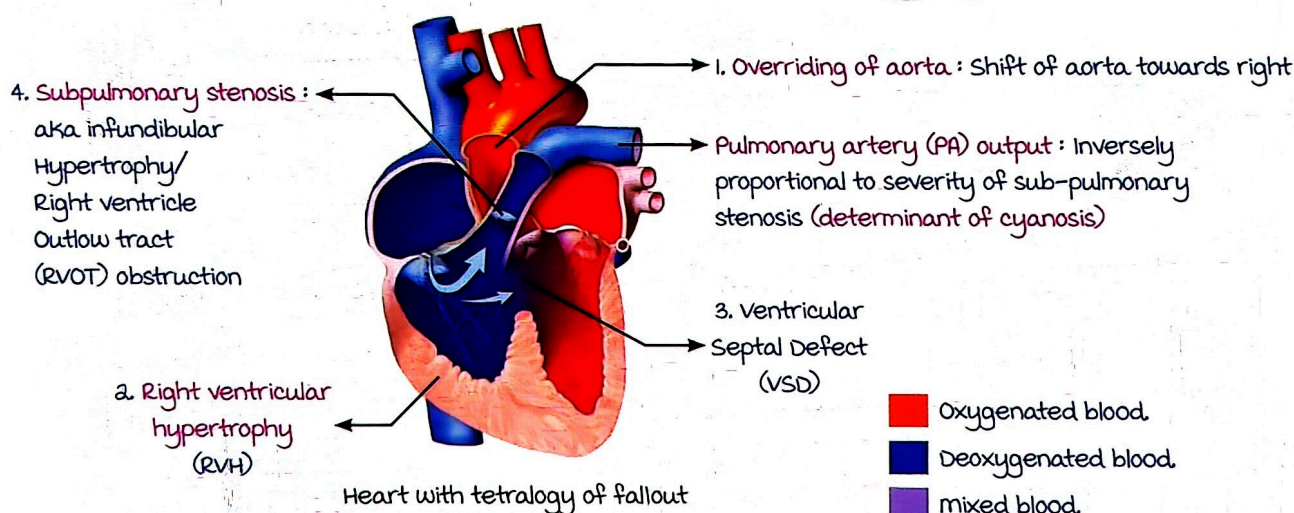
## Classification :



## Tetralogy of Fallot

00:01:34

### 4 components :



### Clinical Features :

1. Cyanosis : Onset within 4 months after birth (varies with severity of sub-pulmonary stenosis).

2. Single S2 heart sound : {

- A2 heard
- P2 not heard d/t stenosis.

3. murmurs :

- Ejection systolic murmur heard in pulmonary area (Disappears during cyanotic spell : Surest sign of TOF).

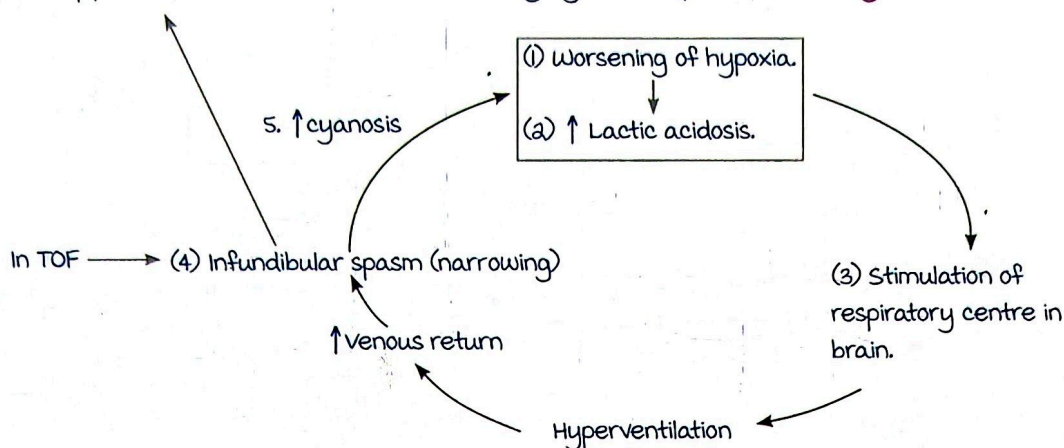
Feedback

----- Active space -----

- VSD is non-restrictive → No turbulent flow → **No pansystolic murmur.**  
(Free flow across VSD)

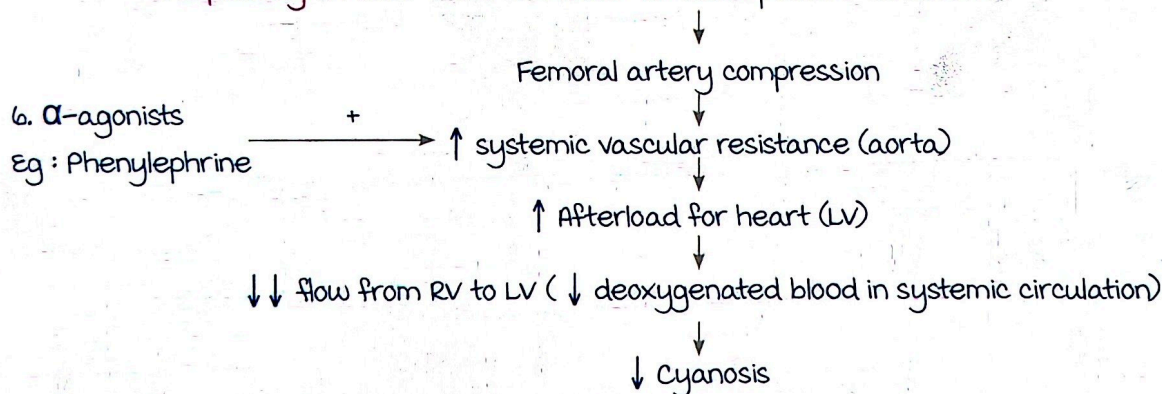
**4. Cyanotic spell:** ↑ severity of cyanosis.

Disappearance of shunt murmur during cyanotic spell: **Surest sign of TOF**



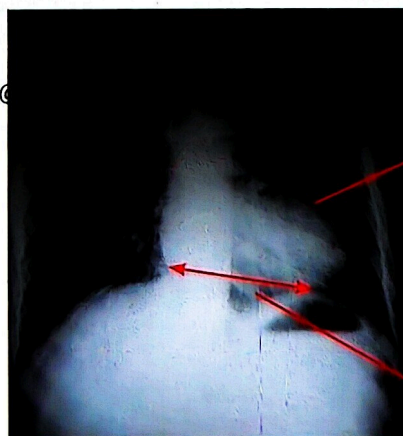
• management:

1.  $O_2$  inhalation.
2. Sodium bicarbonate.
3. morphine: To depress respiratory centre.
4.  $\beta$ -blockers.
5. **Squatting** (In older children)/**knee-to-chest** posture (In infants).



**X-ray findings:**

khanirfan0392@



Boot shaped heart

1. RVH
2. uplifting of apex

**Surgical treatment :**

Definitive : VSD repair + correction of stenosis (Performed < 3 months).

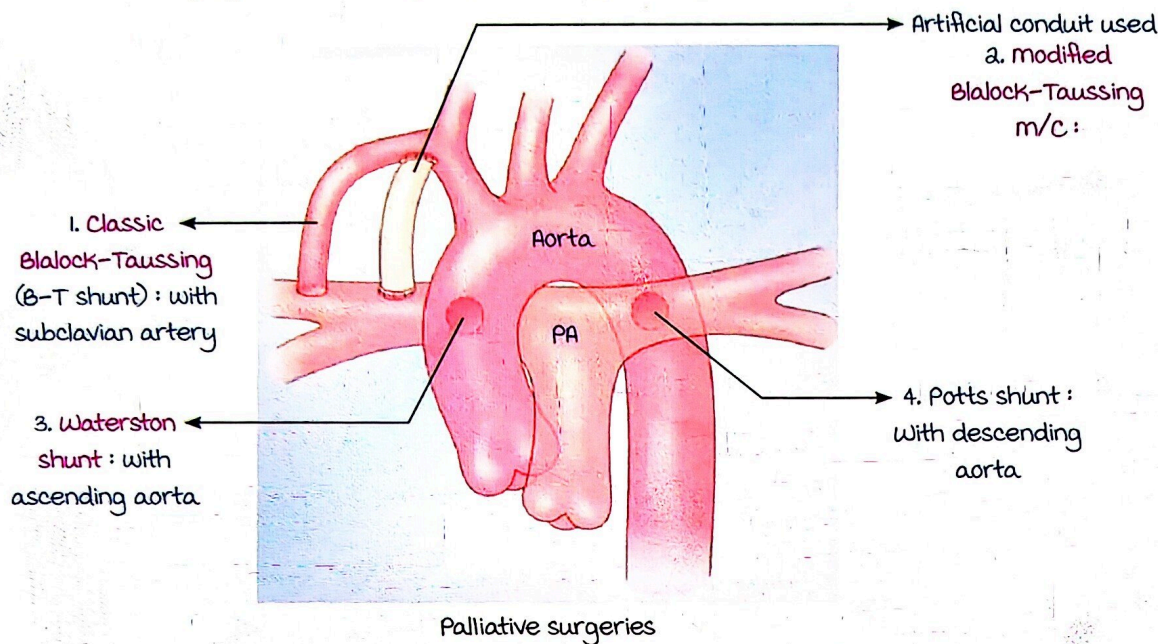
Palliative : Shunt surgery → Improve pulmonary blood flow → Improve cyanosis :

- In delayed presentations.

- C/I to definitive surgery.

method : Shunting systemic artery to pulmonary artery.

----- Active space -----

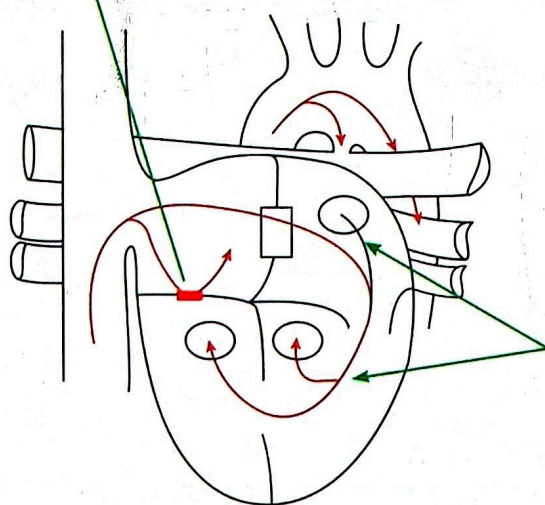


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00:20:21

**Tricuspid Atresia****Pathophysiology :**

Atresia of tricuspid valve → ↓ blood flow to right ventricle → ↓ blood flow to PA → Cyanosis.



a : Patent foramen ovale (PFO) or ASD :

Overload of blood to LA and LV

Hypertrophy of left heart

----- Active space -----

**ECG findings :**

Left-axis deviation seen.

Note : All CHD are diagnosed by ECHO cardiography.

**Treatment :**

Shunt surgery : multiple shunt surgeries are performed in stages.

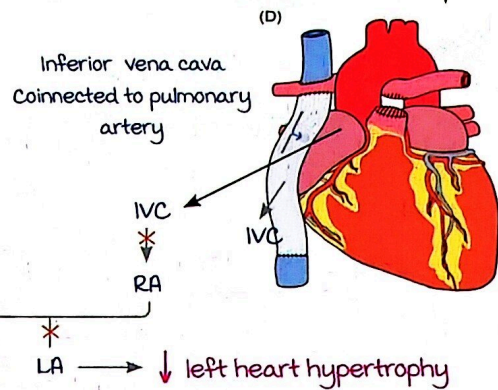
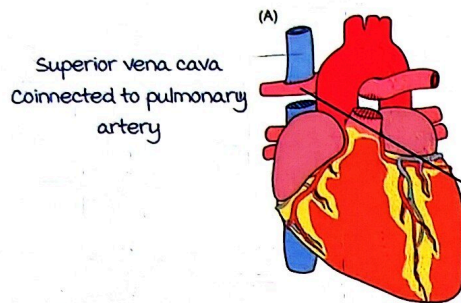
1. modified B-T shunt : Done immediately

↓ followed by

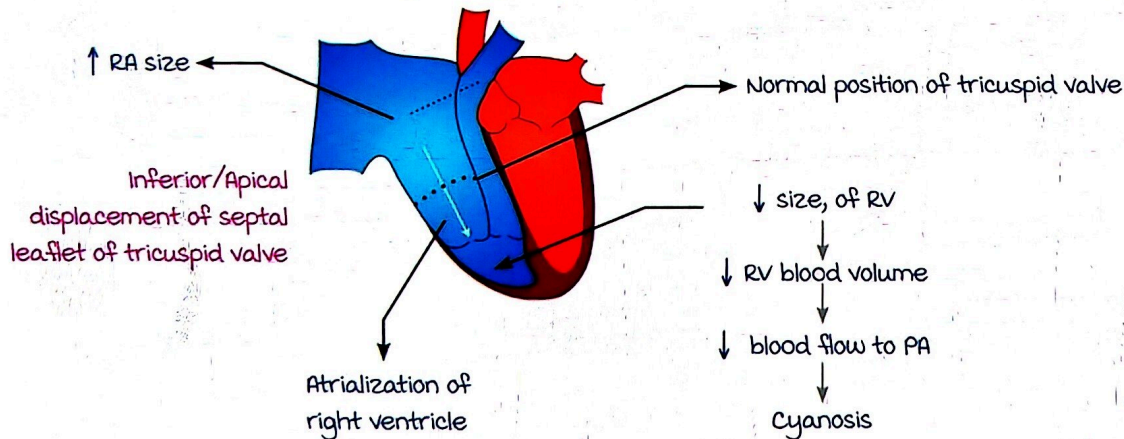
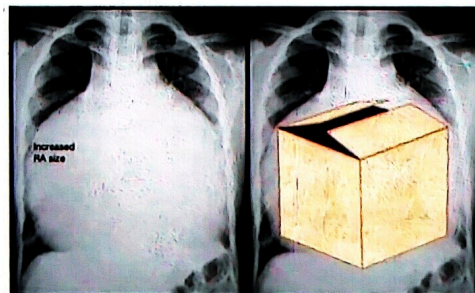
2. Glenn shunt :

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↓ followed by

3. Fontan shunt :

**Ebstein's Anomaly**

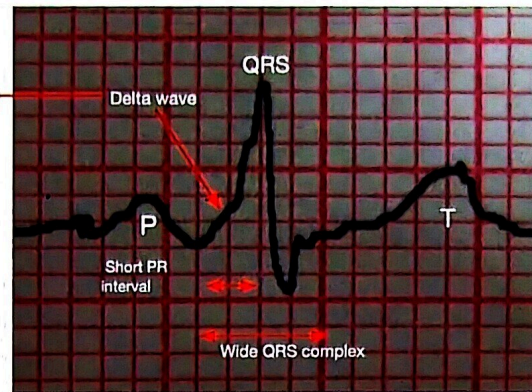
00:27:08

**Pathophysiology :****X-ray findings :**

**Associations :**

1. Lithium exposure.
2. Accessory pathways on right side of heart :

- **Wolff-Parkinson-White syndrome :**  
Abnormal electrical communication between  
RA and RV : Bundle of Kent  
↓  
Early/pre-excitation of RV  
↓  
Slurring of QRS complex



ECG changes

**Treatment :**

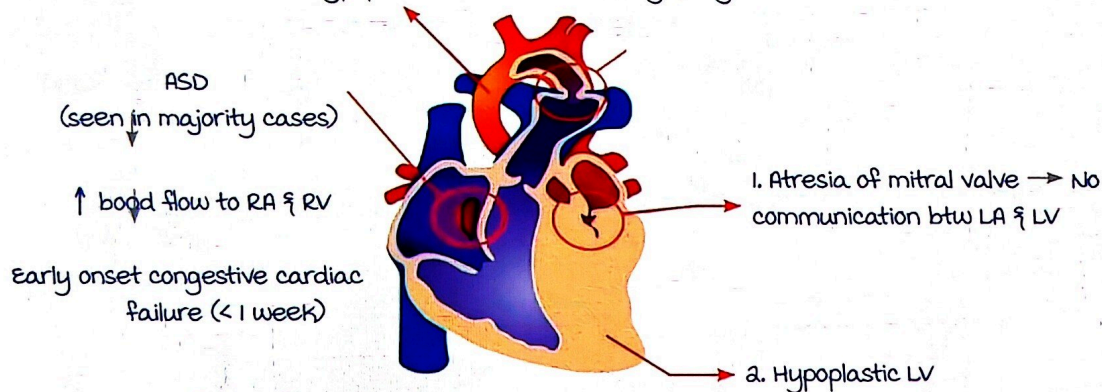
Cone repair of tricuspid valve.  
Return valve to original position.

**Hypoplastic Left Heart Syndrome**

00:33:45

**Pathophysiology :**

3. Hypoplastic aorta → severely ↓ systemic flow → Shock in neonates

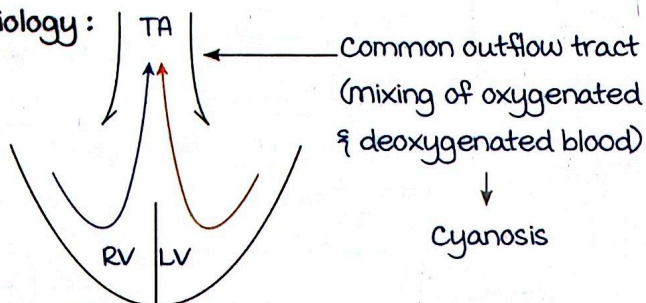
**Treatment :**

Fatal if untreated/delayed treatment.

Norwood procedure : Surgical creation of Neo (new)-aorta to ↑ blood flow.

**Truncus Arteriosus**

00:38:33

**Pathophysiology :**

- Type of conotruncal defect (includes TOF, interrupted aortic arch etc.)  
- Associated with Di-George syndrome.

----- Active space -----

**Treatment:**

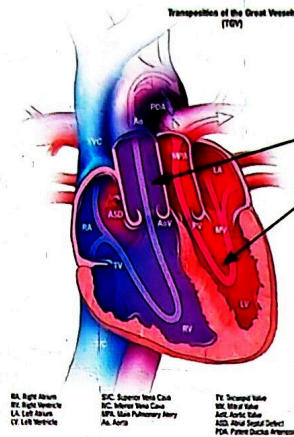
Rastelli's procedure: Truncus arteriosus partitioned → Normal blood outflow.

**Transposition of Great Arteries**

00:41:07

**Pathophysiology:**

1. TGA with intact ventricular septum:

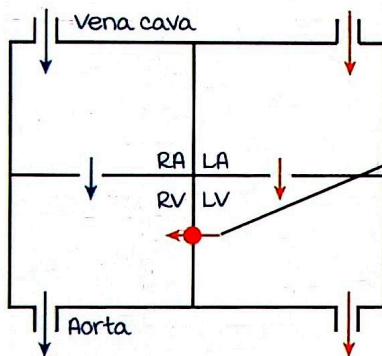


Parallel circulation:  
2 separate blood flow tracts  
for oxygenated &  
de-oxygenated blood

↓  
Early onset cyanosis  
(within few min)

↓  
Fatal

2. TGA with VSD:

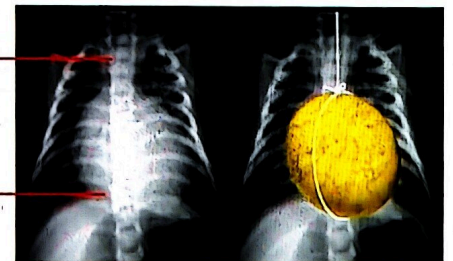


Blood flows from LV to RV  
↓  
Mixing some oxygenated and deoxygenated blood

↓  
Presents with:  
Less intense cyanosis +  
Pansystolic murmur +  
CCF (d/t majority blood flow through parallel pathways)

**X-ray findings:**

String (d/t narrow  
positioning of great  
arteries)



Egg-on-string appearance

**Treatment:**

TGA with intact ventricular septum:

- Emergency condition:
- 1. PGE-1 infusion immediately at birth.  
Keeps ductus arteriosus open: Oxygenated blood from PA to aorta.
- 2. Atrial septotomy: Surgical connection between RA & LA made → ↑ O<sub>2</sub> status.

Definitive treatment

Jatene's arterial switch repair

TGA with VSD:

- Electively (few weeks after birth).

## Total Anomalous Pulmonary Venous Connections

00:49:42

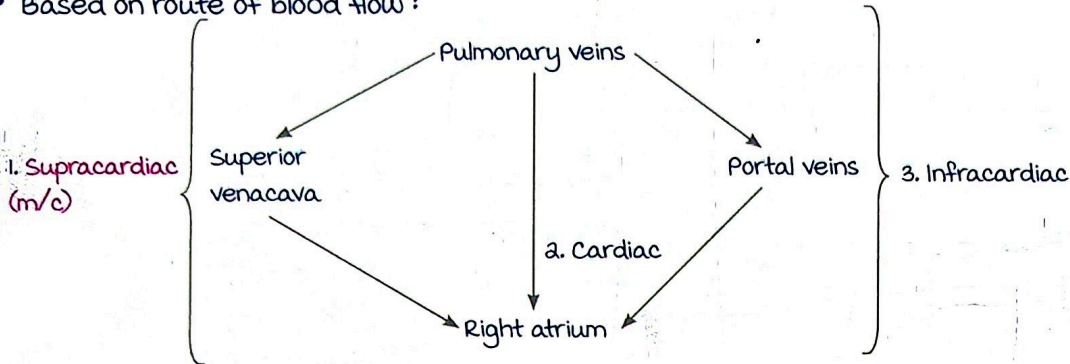
----- Active space -----

### Pathophysiology :

Abnormal connections between PV & RA.

### Types :

- Based on route of blood flow :



### Presentation :

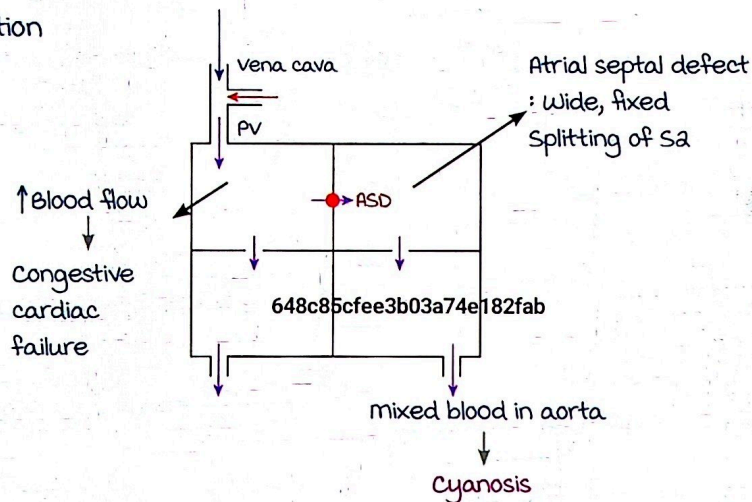
Obstructive type :

Pulmonary venous obstruction

Congestion of lungs

Severe cyanosis

- Non-obstructive type (m/c) : Equal O<sub>2</sub> saturation in all heart chambers.



### X-ray findings :



Supracardiac TAPVC :  
Snowman appearance/Figure of 8  
appearance (d/t ↑ SVC size)